Causality - SoSe 2025

Restricted Models

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So Far

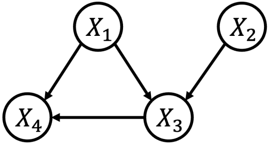
Last lecture, we discussed the following concepts

- Correctness of GS
- Overview on Causal Discovery Frameworks
- SGS and the PC algorithm

So Far

Consider the DAG $\mathcal{G}$ on the right and suppose that:

- We are given observations of variables (X_1, X_2, X_3, X_4)
- The assumptions of the PC algorithm are fulfilled
- We have access to an independence oracle (perfect independence test)



TRUE or **FALSE**?

- If the data come from $\mathcal{G}$, the edge between X_1 and X_2 will be removed first in the skeleton phase.
- The Markov equivalence class of $\mathcal{G}$ only contains one DAG.

Outline

Today we discuss how to distinguish cause from effect by making assumptions about the SCM. We discuss:

- General non-identifiability results
- Identifiability for linear SCMs
- Causal orderings and linear ICA
- Additive noise models
- Regression with subsequent independence testing (RESIT)

Causal Discovery: Overview

Constraint-Based

- Measure **conditional independencies**
- Search for graph which **satisfies** them

Examples

SGS, PC

Identifies

CPDAG

(Markov equivalence class)

Score-Based

- **Score** models based on data fit
- Large search space, greedy approach

Examples

GES

Identifies

CPDAG

(Markov equivalence class)

Functional/Restricted

- **Additional assumptions** on model class, noise distribution, etc.

Examples

LiNGAM, RESIT, CAM

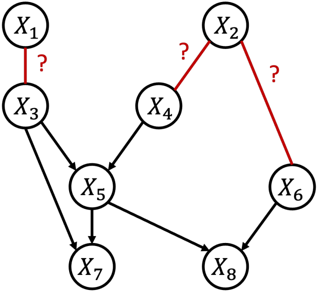
Identifies

DAG

(Usually: True model)

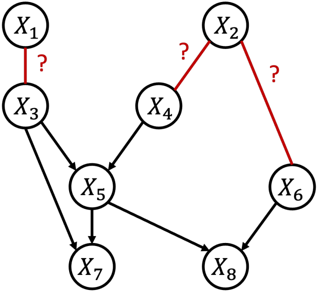
Beyond the Markov Equivalence Class

How can we go *beyond* the Markov equivalence class?



Beyond the Markov Equivalence Class

How can we *identify the direction* of the **red edges**?



Beyond the Markov Equivalence Class

Simply put, *can we distinguish cause from effect?*
That is, distinguish $X \rightarrow Y$ from $Y \rightarrow X$?

- Based on the assumptions we know **so far** (Markov, causal minimality, faithfulness), **no**!
- There is **no v-structure** that we can exploit!
- Both graphs are **Markov equivalent**!

Bi-Variate Models (via SCMs)

Consider an SCM of the form

$$\begin{aligned} X &:= N_X \\ Y &:= f_Y(X, N_Y) \qquad N_X \perp\!\!\!\perp N_Y \end{aligned}$$

which entails a graph $X \rightarrow Y$ and a joint PDF p_{XY}.

Given the joint p_{XY}, can we infer the SCM $X \rightarrow Y$ and exclude $Y \rightarrow X$?

Proposition For any joint PDF p_{XY} there exists an SCM encoding the graph $X \rightarrow Y$, i.e., there exists N_X, N_Y, f_Y, s.t.

$$X := N_X \quad Y := f_Y(X, N_Y) \quad N_X \perp\!\!\!\perp N_Y \ .$$

Thus, for any p_{XY}, there exists a SCM $X \rightarrow Y$ but also an SCM $Y \rightarrow X$. Hence, **no!** We cannot distinguish those graphs based on p_{XY}.

Bi-Variate Models (via SCMs)

Proposition For any joint PDF p_{XY} there exists an SCM encoding the graph $X \to Y$, i.e., there exists N_X, N_Y, f_Y, s.t.

$$X := N_X \quad Y := f_Y(X, N_Y) \quad N_X \perp\!\!\!\perp N_Y \; .$$

Proof:

Define the conditional CDF

$$F_{Y|x}(y) = P(Y \leq y | X = x) = \int_{-\infty}^{y} p(y' | X = x) dy' \; .$$

Let f_Y be the inverse, i.e., the corresponding quantile function, that is

$$f_Y(x, n_y) = F_{Y|x}^{-1}(n_y) = \inf\{y \in \mathbb{R} \mid F_{Y|x}(y) \geq n_y\} \; .$$

define $N_X = X$. Futher, let N_Y to be uniformly distributed on $[0, 1]$ independent of X. Then N_X, N_Y, f_Y define an SCM with joint PDF p_{XY}. ■

Bi-Variate Models

Conclusion

To decide between $X \to Y$ and $Y \to X$ from the joint PDF p_{XY} (or iid observations of p_{XY}), *we need additional assumptions!*

Additional Assumptions

- The **mechanism** and the **cause** should be **independent**
- **More formally**: The **conditional** $p_{X|Y}$ and the **marginal** p_X of the cause X should be somehow **independent**.

Phrased as an assumption, the above is known as the **principle of independent mechanisms**. We will discuss some cases in which it holds.

LiNGAM

Linear Models with Additive Noise

Definition A PDF p_{XY} admits a **linear additive model (LAM)** $X \to Y$, if there exists α and N_Y, s.t.

$$Y := \alpha X + N_Y \quad N_Y \perp\!\!\!\perp X$$

with continuous RVs X, N_Y, Y.

Independence assumption?

- Assuming $N_Y \perp\!\!\!\perp X$ "suggests" some sort of independence between $p_{Y|X}$ and p_X since $p_{Y|x}(y|x) = p_{N_Y}(y - \alpha x)$

Linear Models with Additive Noise

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$$Y := \alpha X + N_Y \quad N_Y \perp\!\!\!\perp X$$

with continuous RVs X, N_Y, Y.

Theorem p_{XY} admits a LAM. Then **1.** and **2.** are **equivalent**:

1. There **exists a backward LAM**, i.e, there exists β and a RV N_X, s.t.

$$X := \beta Y + N_X \quad N_X \perp\!\!\!\perp Y \ .$$

2. N_Y and X are **Gaussian**.

The SCM $X \to Y$ is **identifiable** if X or N_Y are **non-Gaussian**!

Proof

2. $\implies$ 1.: Regress X on Y to get β. Linear combinations of Gaussians are Gaussian, i.e., Y is Gaussian, and the residuals $N_X = X - \beta Y$ are Gaussian (and always uncorrelated from Y in linear regression); hence independent.

1. $\implies$ 2.: uses the **Darmois-Skitovič** theorem:

Let A, B be independent RVs, if $\exists \alpha_1, \alpha_2, \beta_1, \beta_2 \neq 0$, $\alpha_i \neq \beta_i$, s.t.

$$\alpha_1 A + \alpha_2 B \perp\!\!\!\perp \beta_1 A + \beta_2 B$$

then A and B are Gaussian distributed.

We know that X and N_Y are independent, so they take the roles of A and B:

$$Y = \alpha X + N_Y$$
$$N_X = X - \beta(\alpha X + N_Y) = (1 - \beta\alpha)X - \beta N_Y$$

i) $\beta \neq 0$ and $(1 - \beta\alpha) \neq 0$: We can apply the theorem above.
ii) $\beta = 0$: We get a contradiction, i.e., $\alpha X + N_Y \perp\!\!\!\perp X$.
iii) $(1 - \beta\alpha) = 0$: We get a contradiction, i.e., $\alpha X + N_Y \perp\!\!\!\perp \beta N_Y$. ■

LiNGAM

Definition p_{XY} admits a **linear non-Gaussian additive noise model** (LiNGAM) $X \to Y$, if there exists α and N_Y, s.t.

$$Y := \alpha X + N_Y \quad N_Y \perp\!\!\!\perp X$$

with continuous RVs X, N_Y, Y, where X **or** N_Y **are non-Gaussian**.

- If p_{XY} admits LiNGAM, then $X \to Y$ is **identifiable**!
  (can be derived based on the previous theorem)
- This result has been derived by Shimizu et al. in 2006

LiNGAM Example

Given the SCM $X \to Y$:

$$Y := 1/2X + N_Y \quad X \perp\!\!\!\perp N_Y$$

with X and N_Y are uniformly distributed (**non-Gaussian**).

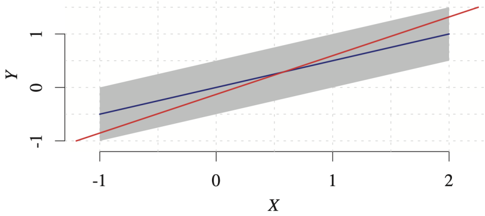


Figure 1: Source: ECI book

The blue line is the forward model, and the red line is the least squares fit minimizing $E[X - \beta Y - \gamma]$. The gray area is the support of the joint distribution. **Note that** the size of the support of the estimated N_X would differ for different values of Y.

LiNGAM Example (Empirical)

Given the SCM $X \to Y$:

$$Y := 1/2X + N_Y \quad X \perp\!\!\!\perp N_Y$$

with X and N_Y are uniformly distributed (**non-Gaussian**).

Observations:

- **Forward model**: independent residuals

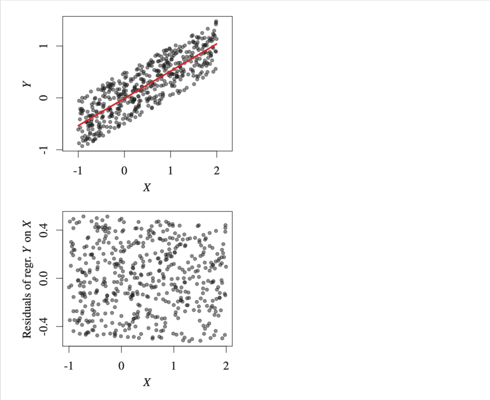


Figure 2: Source: ECI book (colors inverted compared to the previous figure)

LiNGAM Example (Empirical)

Given the SCM $X \to Y$:

$$Y := 1/2X + N_Y \quad X \perp\!\!\!\perp N_Y$$

with X and N_Y are uniformly distributed (**non-Gaussian**).

Observations:

- **Forward model**: independent residuals
- **Backward model**: *uncorrelated* but **not independent** residuals

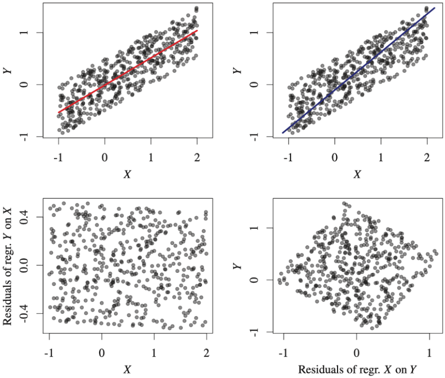


Figure 3: Source: ECI book

Here, we can identify $X \to Y$ with a **nonlinear** independence test.

Beyond Pairs

If $\boldsymbol{X_V}$[1] (simply $\boldsymbol{X}$) contains $d \geq 2$ variables, we can write a linear SCM as

$$\boldsymbol{X} := \boldsymbol{BX} + \boldsymbol{N} \quad \text{with } \boldsymbol{B} \in \mathbb{R}^{d \times d}, \boldsymbol{X} \in \mathbb{R}^d, \boldsymbol{N} \in \mathbb{R}^d \tag{1}$$

where $N_i \perp\!\!\!\perp N_j$ for $i \neq j$.

For $p = 2$ with $X_1 \to X_2$, Eq. (1) simplifies to

$$\begin{bmatrix} X_1 \\ X_2 \end{bmatrix} := \begin{bmatrix} 0 & 0 \\ \alpha & 0 \end{bmatrix} \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} + \begin{bmatrix} N_1 \\ N_2 \end{bmatrix}$$

- Due to acyclicity (& no self-loops) the diagonal of $\boldsymbol{B}$ is zero
- Permuting the order of variables we can (using a **causal ordering**) make $\boldsymbol{B}$ strictly **lower triangular**

[1] $\boldsymbol{X_V}$ are generated through a SCM which defines a DAG $\mathcal{G}$ over nodes $\boldsymbol{V}$.

Causal Ordering

Definition Given a DAG $\mathcal{G}$ with d nodes, we say that a bijective mapping π

$$\pi : \{1, \ldots, d\} \to \{1, \ldots, d\}$$

is a **causal ordering** of the variables if it satisfies

$$\pi(i) < \pi(j) \text{ if } j \in \mathrm{De}(i) \ .$$

- Due to acyclicity, there is always a causal odering (but it is not necessarily unique)
- For example, for the graph $1 \to 3 \leftarrow 2$, the identity would be a valid ordering, however, also an ordering that swaps nodes $1, 2$ is valid, i.e,

$$\pi(1) = 2, \ \pi(2) = 1, \ \pi(3) = 3$$

Multivariate LiNGAM

Goal: Based on n iid observations of $\boldsymbol{X_V}$ estimate $\boldsymbol{B}$. Since non-zeros in $\boldsymbol{B}$ are edges in the DAG, we also learn the DAG.

Estimation methods:

- ICA-LiNGAM
  - Can exploit fast **independent component analysis** (ICA) methods, but has issues with local minima
- DirectLiNGAM
  - Guaranteed convergence for $n \to \infty$; iteratively finds source nodes to get to a causal order

Independent Component Analysis: The Cocktail Party

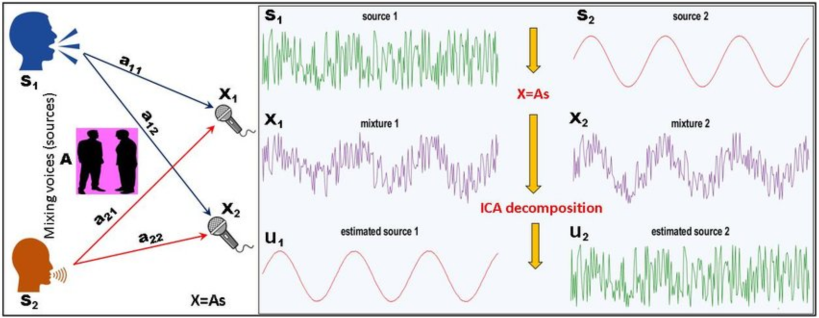


Figure 4: Source: Hassan, Goussev. *Fusion of multi-physics data with machine learning independent component analysis (ICA) improves detection of geological features*. 2019

Independent Component Analysis

Independent component analysis (ICA):

- *non-Gaussian variant of factor analysis*
- *cocktail party problem (disentangle who is talking)*

Definition The ICA model is defined as

$$\boldsymbol{X} := \boldsymbol{A}\boldsymbol{S}$$

- $\boldsymbol{X} \in \mathbb{R}^d$: observed variables
- $\boldsymbol{S} \in \mathbb{R}^d$: mutually **independent**, continuous latent (unobserved) **non-Gaussian** variables (**sources**)
- $\boldsymbol{A} \in \mathbb{R}^{d \times d}$: unobserved full-rank **mixing-matrix**

If $\boldsymbol{S}$ is non-Gaussian, then $\boldsymbol{A}$ is identifiable up to permutation, scaling and sign of the columns!

ICA Intuition for Identifiability

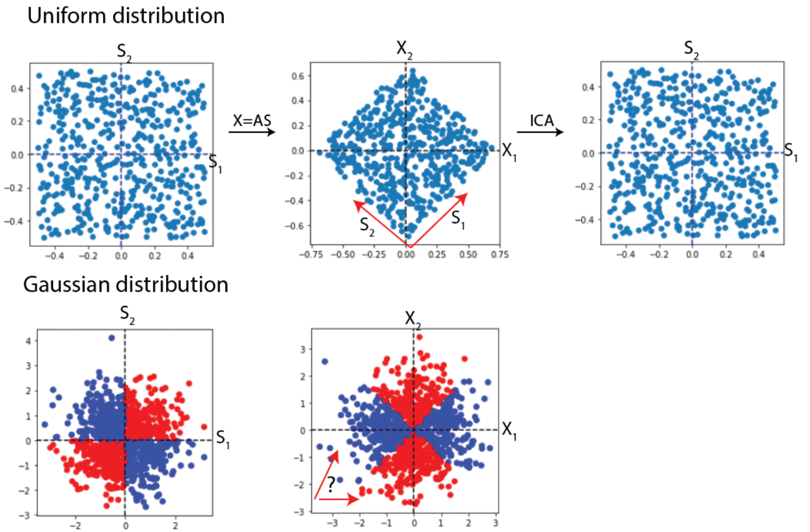


Figure 5: Top: The independent compontes S_1 and S_2 can be identified (non-Gaussian data). Bottom: For Gaussian data, we cannot find an $\boldsymbol{S}$ that is unique up to permutation, sclaing and the sign of the columns.

Multivariate LiNGAM

We can re-write a linear SCM as follows

$$\begin{aligned} \boldsymbol{X} &= \boldsymbol{B}\boldsymbol{X} + \boldsymbol{N} \\ (\boldsymbol{I} - \boldsymbol{B})\boldsymbol{X} &= \boldsymbol{N} \\ \boldsymbol{X} &= (\boldsymbol{I} - \boldsymbol{B})^{-1}\boldsymbol{N} \end{aligned}$$

- LiNGAM is an instance of the ICA model $\boldsymbol{X} = \boldsymbol{A}\boldsymbol{S}$, with $(\boldsymbol{I} - \boldsymbol{B})^{-1} = A$ and $\boldsymbol{N} = \boldsymbol{S}$
- ICA identifies $\boldsymbol{A}$ up to permutation, scaling and sign of the columns
- We can exploit futher properties of $\boldsymbol{B}$: **zero on the diagonal**, and **strictly lower triangular**

ICA-LiNGAM

ICA-LiNGAM Algorithm:

1. Estimate $\boldsymbol{W} = \boldsymbol{A}^{-1} = (\boldsymbol{I} - \boldsymbol{B})$ from iid observations of $\boldsymbol{X_V}$ via ICA (up to permutation, scaling and sign of the columns of *the true* $\boldsymbol{A}$ / rows *of the true* $\boldsymbol{W}$)
2. Find unique row permutation $\tilde{\boldsymbol{W}}$ of $\boldsymbol{W}$ without zeros on the diagonal
   - Linear assignement problem (found by minimizing $\sum_i 1/|\tilde{\boldsymbol{W}}_{ii}|$)
3. Get $\tilde{\boldsymbol{W}}'$ by dividing each row of $\tilde{\boldsymbol{W}}$ by its diagonal element
4. Compute $\hat{\boldsymbol{B}} = \boldsymbol{I} - \tilde{\boldsymbol{W}}'$
5. Find causal order by permuting $\hat{\boldsymbol{B}}$ such that it is as close as possible to a strictly lower triangular matrix.
6. (Optional) Apply pruning based on sparse regression of significance testing to get a sparse DAG.

Summary LiNGAM

- Identifiability can be proven similar to the bi-variate case based on the multivariate version of the Darmois-Skitovič theorem
- **Non-Gaussianity** is crucial for LiNGAM and ICA
- LiNGAM **does not require faithfulness**
- LiNGAM implicitely requires **structural minimality**, i.e., no coefficient should be zero (Remark 6.6, lecture 4)
- We can learn the **full DAG** with ICA-LiNGAM

What about more complicated functions?

Nonlinear Additive Noise Models

Nonlinear Additive Noise Models

Definition p_{XY} admits an **additive noise model (ANM)** $X \to Y$, if there exists a function f_Y, and RVs X, N_Y, s.t.

$$Y := f_Y(X) + N_Y \quad X \perp\!\!\!\perp N_Y \ .$$

Independence assumption?

- Again, assuming $N_Y \perp\!\!\!\perp X$ "suggests" some sort of independence between $p_{Y|X}$ and p_X

Nonlinear Additive Noise Models

Definition p_{XY} admits an **additive noise model (ANM)** $X \to Y$, if there exists a function f_Y, and RVs X, N_Y, s.t.

$$Y := f_Y(X) + N_Y \quad X \perp\!\!\!\perp N_Y \; .$$

Theorem If p_{XY} admits a **Gaussian** ANM $X \to Y$ (N_Y and X are Gaussian), then **1.** implies **2.**:

1. p_{XY} admits an ANM $Y \to X$
2. f_Y is linear.

- Simply put: Gaussian ANMs are *identifiable* for **nonlinear** functions
- A more general theorem that describes all cases for which ANMs are non-identifiable (i.e., including non-Gaussian distributions) is provided as Theorem 4.6 in the ECI book

Example: Non-Identifiable Distributions

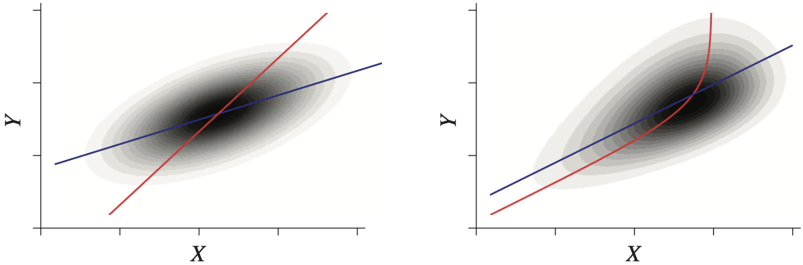


Figure 6: Joint density over X and Y for two non-identifiable examples. The left panel shows the linear Gaussian case and the right panel shows a slightly more complicated example, with "fine-tuned" parameters for function, input, and noise distribution (the latter plot is based on kernel density estimation). The blue function f_Y corresponds to the forward model, and the red function to the backward model f_X.

RESIT

REgression with Subsequent Independence Testing (RESIT)

(Special case of RESIT for 2 variables)

1. Regress Y on X using some (possibly nonlinear) regression technique (obtaining $\hat{f}_Y$)
2. Test if the **residuals** $Y - \hat{f}_Y(X)$ are **independent** of X
3. Repeat the procedure for the inverse direction
4. If independence is not rejected for one direction and rejected for the other, infer the former as the causal direction
   - In practice: Infer the direction with the higher p-value (less evidence to reject independence)

Example for RESIT[2]

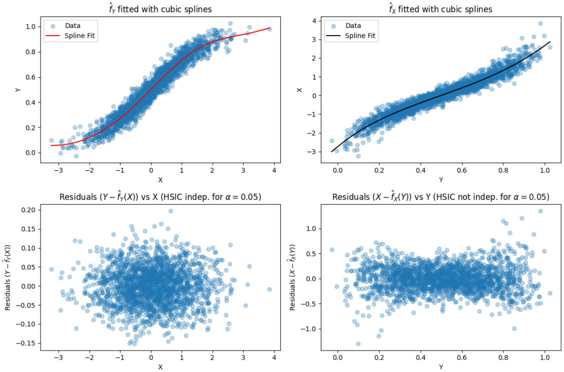


Figure 7: The true model is $X \rightarrow Y$ (Gaussian ANM). Shown are the fits for the **forward model** in red and the **backward model** in black, as well as the residuals and the result of an independence test for testing the independence of residuals and potential cause (HSIC, which is a kernel independence test).

[2] The Python code to reproduce the plots will be provided.

Multivariate RESIT

Definition $p_{\boldsymbol{X}}$ admits a multivariate ANM, if the structural assignment for each node i can be written as

$$X_i := f_i(\boldsymbol{X}_{\mathrm{Pa}(i)}) + N_i$$

where all N_i are pairwise independent.

Multivariate RESIT:

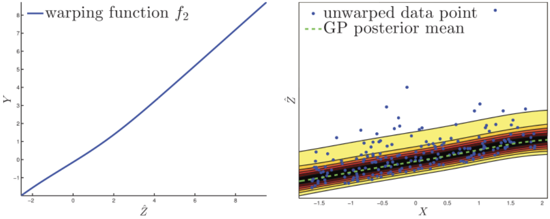
- **Key idea**: Iteratively identify **sink nodes** (nodes without children)
  - Regress each node based on all potential parents
  - Identify the node (sink) for which the residuals are **most independent** of the potential causes
- Requires **causal minimality** (faithfulness is not needed)
- Requires a **suitable regression method** and **independence test**

Is "nonlinear additive" the limit?

Post-Nonlinear Additive Noise[3]

Definition p_{XY} admits an **post-nonlinear additive noise (PNL) model** $X \rightarrow Y$, if there exist nonlinear f_1, f_2, and RVs X, N, s.t.

$$Y := f_2(\underbrace{f_1(X) + N}_{\text{ANM}}) \quad X \perp\!\!\!\perp N \ .$$



(a) Estimated PNL function f_2 (b) Estimated f_1

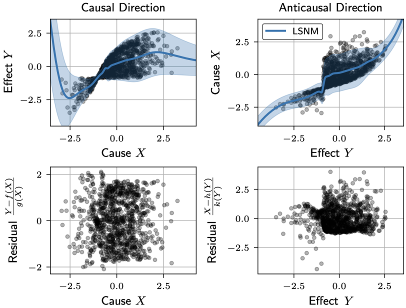
Figure 8: Zhang et al. (2015) *On Estimation of Functional Causal Models: General Results and Application to the PNL Causal Model*

[3]Identifiability is disussed in (Zhang and Hyvärinen, 2009)

Location-Scale Noise Models[4]

Definition p_{XY} admits an **location-scale noise model (LSMN)** $X \to Y$, if there exist nonlinear functions f, g (g is strictly positive), and random variables X, N, s.t.

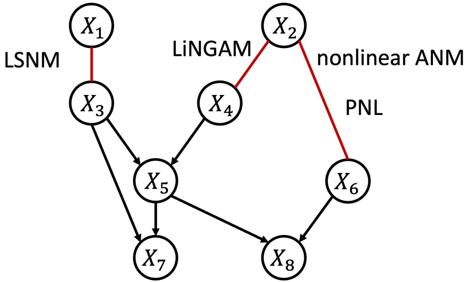
$$Y := f(X) + g(X)N \quad X \perp\!\!\!\perp N\,.$$



[4] Identifiability is discussed in (Immer, et al., 2023)

Summary

How can we direct the **red edges**? - **Restricting the SCM!**



Note

- Each class has **non-identifiable cases** (e.g., linear Gaussian)
- All of those results are derived under **causal sufficiency**
- **Estimating PNL and LSNM** models efficiently is challenging

Credits

- **Recommended reading** ECI book Chapter 4.
- This course *builds upon and extends* the material from *"Causality" (WiSe 2022/23)* that was kindly provided by **Stefan Harmeling**
- This lecture *builds upon and extends* the slides of the course *"Causality" (2021)* by Christina Heinze-Deml

Thank you!